Day 2 Notes: DSA Optimization & Machine Learning Core Concepts

Date: October 27, 2026 **Topics Covered:** DSA (Hash Maps), Machine Learning (Supervised vs. Unsupervised).

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1. DSA: Optimizing "Two Sum"

The Goal

Optimize the brute-force O(n²) solution from Day 1 to an O(n) solution.

The Concept: Hash Maps

A hash map (Python dictionary) allows us to store data as **key-value pairs** and look them up in O(1) average time. Instead of searching the whole list repeatedly, we ask the dictionary whether we've seen a number before.

The Logic

Loop through the list of numbers.

Calculate the `complement` (`target - current_number`).

Check whether the `complement` is already in our dictionary.

If yes, return the indices of the current number and the complement.

If no, add the current number and its index to the dictionary.

[Insert image here: Screenshot of your LeetCode green checkmark]

The Code

```
class Solution(object):
    def twoSum(self, nums, target):
        num_map = {}
        for i, num in enumerate(nums):
            complement = target - num
            if complement in num_map:
                return [num_map[complement], i]
            num_map[num] = i
```

Complexity: O(n) average time and O(n) extra space.

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2. Machine Learning: Core Concepts

Machine learning is often introduced through two broad categories based on the kind of data available and how the model learns.

A. Supervised Learning

The model learns from **labeled data** (input + correct output). We act as the teacher by providing the answers.

1. Regression

Definition: Predicting a continuous numerical value, such as price, temperature, or age.

Types of regression:

- **Linear Regression:** Fits a straight line to model the relationship between variables.

- **Multiple Regression:** Uses more than one predictor variable to predict an outcome.
- **Polynomial Regression:** Models curved relationships by using polynomial features.
- **Ridge/Lasso Regression:** Regularization techniques that help reduce overfitting. Ridge uses an L2 penalty, while Lasso uses an L1 penalty and can shrink some coefficients to zero.

Example: Predicting a house's price based on square footage, location, and number of bedrooms.

2. Classification

Definition: Predicting a discrete category or class, such as Yes/No or Spam/Not Spam.

Algorithms: Logistic Regression (used for classification despite its name), Decision Trees, Random Forests, and Support Vector Machines (SVMs).

Example: Predicting whether a tumor is "Malignant" or "Benign," or whether an email is spam.

B. Unsupervised Learning

The model learns from **unlabeled data** (input without target labels). It looks for patterns, structures, or relationships without an answer key.

1. Dimensionality Reduction

Definition: Reducing the number of features (columns) in a dataset while retaining as much useful information as possible. This can make data easier to visualize and reduce computation.

Example: Reducing a dataset with 100 columns to 3 dimensions so it can be plotted on a 3D graph.

Algorithm: Principal Component Analysis (PCA).

2. Anomaly Detection

Definition: Identifying rare observations or events that differ significantly from the majority of the data.

Example: A credit-card transaction may be flagged as suspicious if it differs from the user's usual spending patterns. A large transaction alone does not necessarily mean fraud.

3. Association Rule Learning

Definition: Discovering rules that describe how items or events occur together in a dataset.

Example: "People who bought X also bought Y," which can support product recommendations or grocery-store layout decisions.

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3. Statistics

Skipped today due to a schedule shift and late start. Will resume tomorrow.

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4. Daily Journal

What went well:

- Successfully optimized Two Sum to O(n) average time.
- Understood the difference between Regression and Classification.
- Switched to a more structured DSA platform.

What was hard:

Managing the schedule after oversleeping. I had to drop Statistics to protect my energy and avoid burnout.

Tomorrow's goal:

Get back to the planned timetable with a 10 AM start and work through the Statistics numericals.

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